

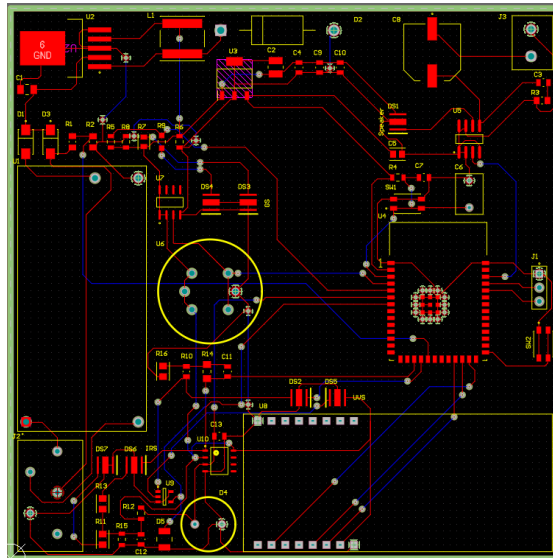
Validation Report

PCB Design Process:

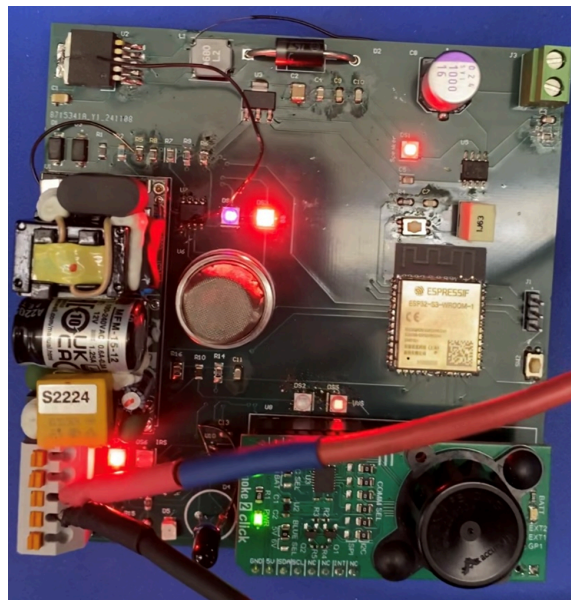
Iteration 1:

Functional, but the alarm speaker was too quiet due to the output signal voltage from the MCU, and the wall-power terminals were located dangerously close together.

Additionally, the IR and UV sensor modules were designed incorrectly, causing parasitics between sensor signals.



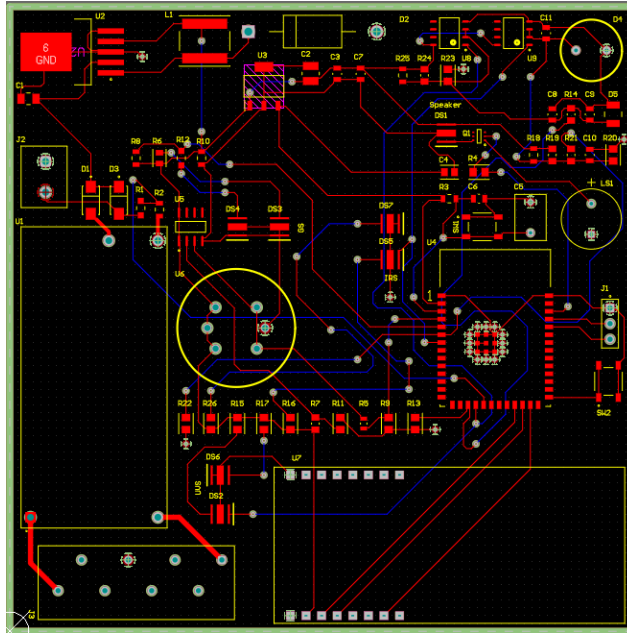
1st Iteration PCB Layout



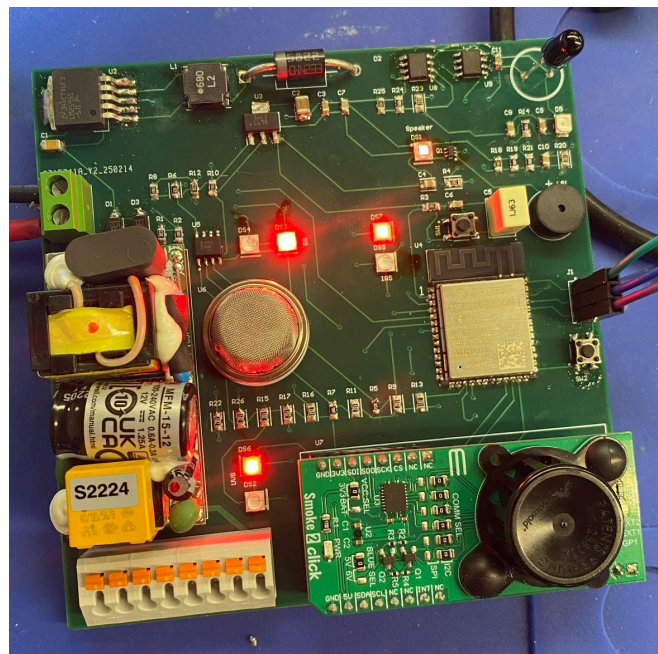
1st Iteration constructed PCB

Iteration 2:

The audible alarm system was changed from a speaker to a buzzer that would be activated by a transistor switch controlled by the MCU. The battery-power connection terminals were also given a separate connector block from the wall-power connection terminals, allowing wall-power to have a dedicated, wider block for safety purposes. Additionally, the IR sensor module was tweaked and the UV sensor module was redesigned to fix the signal calibration and entanglement problems.

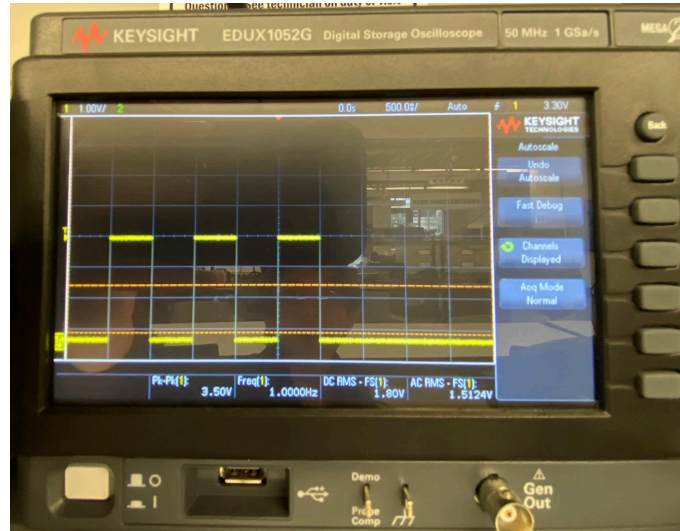


2nd Iteration PCB Layout



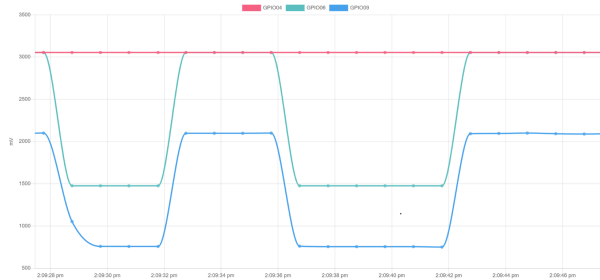
2nd Iteration Constructed PCB

Signal Testing:



Buzzer alarm control signal from MCU as seen in Oscilloscope

Analog Value Display using ESP32



IR and UV signals coupled from 1st PCB

Analog Value Display using ESP32



Gas Sensor readings

Analog Value Display using ESP32

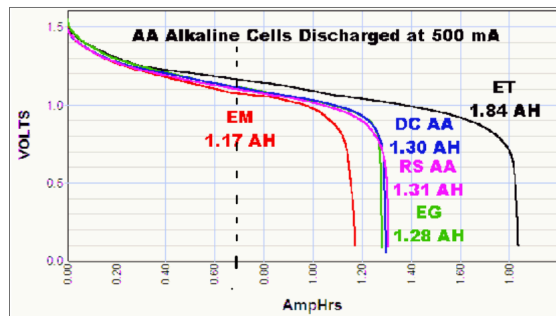


Battery power (test 9V dropped to 7V)

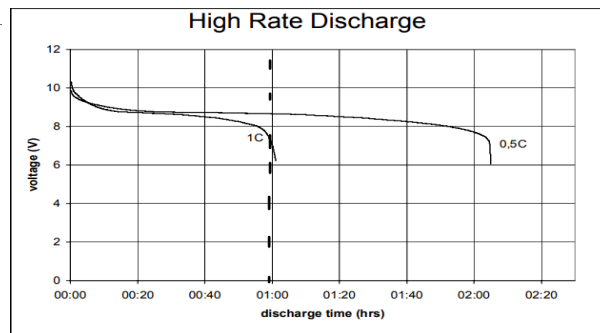
Power Consumption Data:

Power consumption using battery:

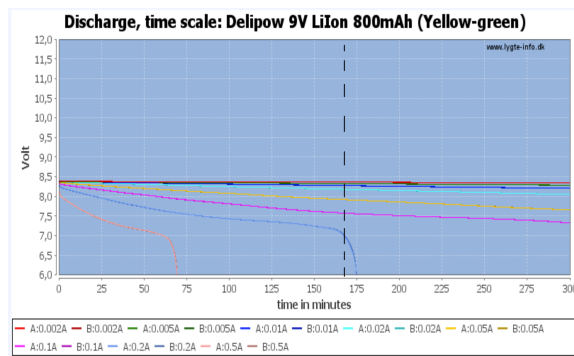
Setup	Voltage (V)	Current (A)	Power (W)
Without Wi-Fi	8.99	0.260	2.35
With Wi-Fi on	9	0.3	2.7



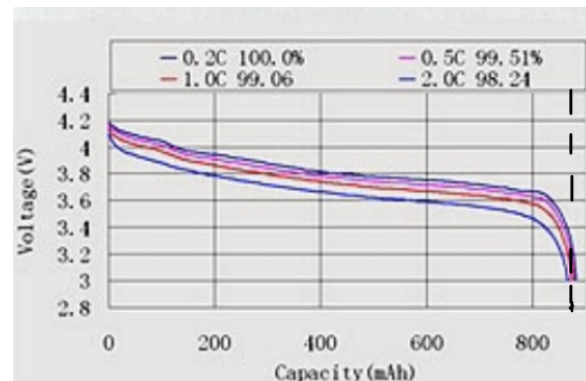
Typical Alkaline battery discharge rate



Typical NiMh battery discharge rate



Typical Lithium-ion battery discharge rate



Typical Lithium-polymer battery discharge rate

<https://www.powerstream.com/AA-tests.htm>

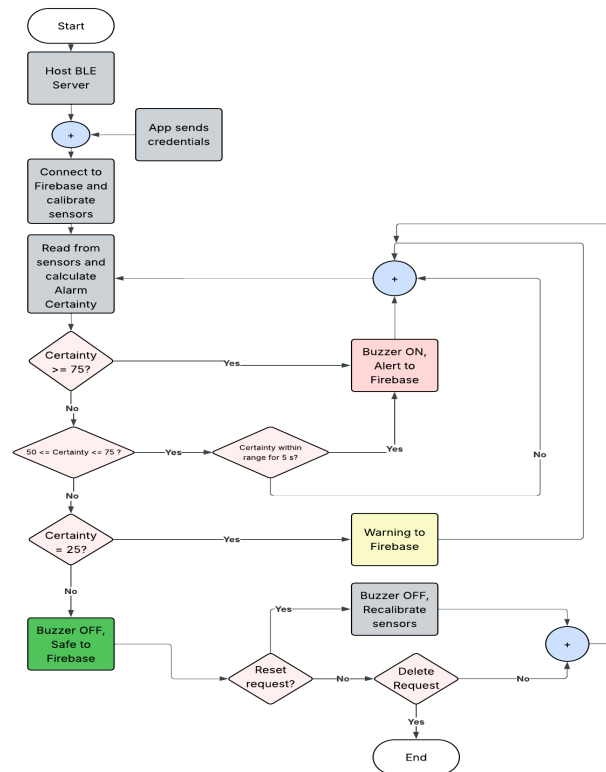
<https://www.farnell.com/datasheets/39693.pdf>

<https://lygte-info.dk/review/batteries9V/Delipow%209V%20LiIon%20800mAh%20%28Yellow-green%29%20UK.html>

<http://www.ibt-power.com/bkencel/Lithium-Po-tech.html>

Battery Chemistry	Expected Run-Time on Battery Power (Hours)
Alkaline	0.7
NiMh	1
Lithium-ion	2.81
Lithium-polymer	2.87

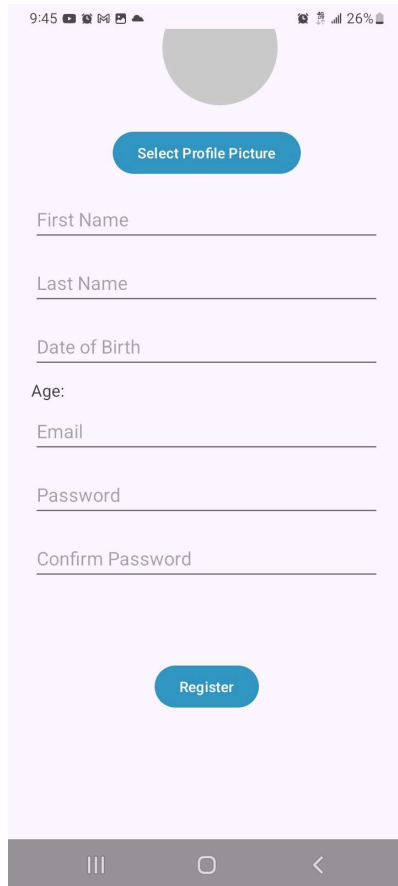
MCU Performance:



High Level Flowchart

Requirement	Specification	Result	Status
Device Setup Time	< 2 s	6.59 s avg.	Delayed
Firebase Push Latency	< 1 s	0.578 s avg.	Pass
Battery Monitoring	Good if > 7.2V	Brownout $\approx$ 6.7 V	Pass - Warning sent at 7.6 V
Alarm Confirmation Logic	$\geq$ 75% to trigger	$\approx$ 99%	Pass
Time to Activate Buzzer upon Detection	< 5 s	< 1 us	Pass
Time to Alert App upon Detection	< 5 s	$\approx$ 1 s	Pass

Application Features:



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Select Profile Picture

First Name

Last Name

Date of Birth

Age:

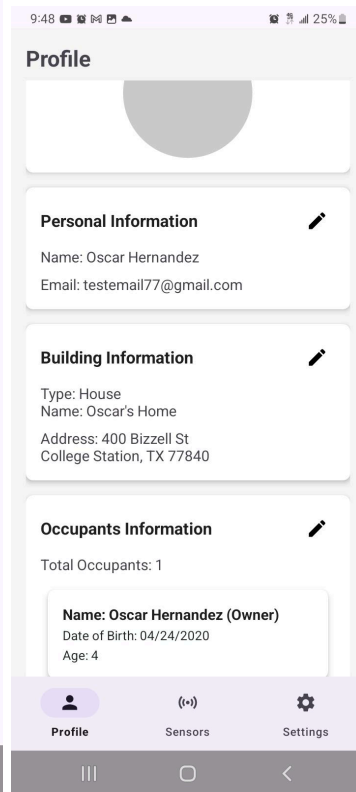
Email

Password

Confirm Password

Register

User Creation



9:48 25%

Profile

Personal Information

Name: Oscar Hernandez
Email: testemail77@gmail.com

Building Information

Type: House
Name: Oscar's Home
Address: 400 Bizzell St
College Station, TX 77840

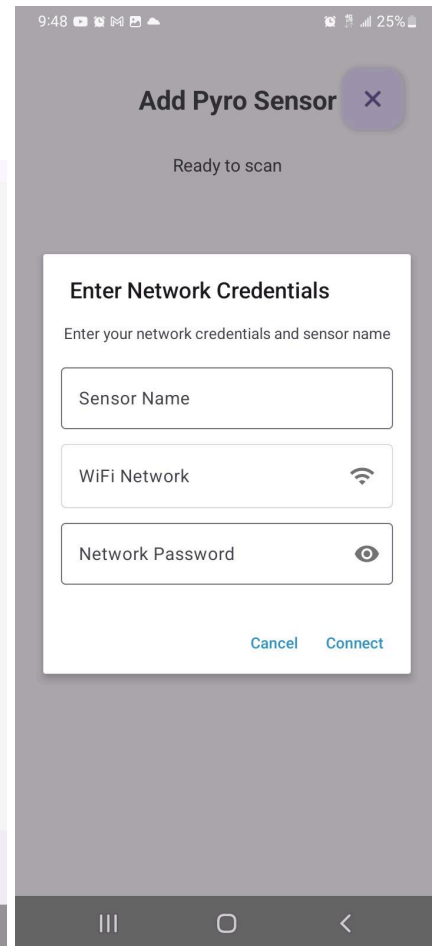
Occupants Information

Total Occupants: 1

Name: Oscar Hernandez (Owner)
Date of Birth: 04/24/2020
Age: 4

Profile Sensors Settings

Profile Edit



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Add Pyro Sensor

Ready to scan

Enter Network Credentials

Enter your network credentials and sensor name

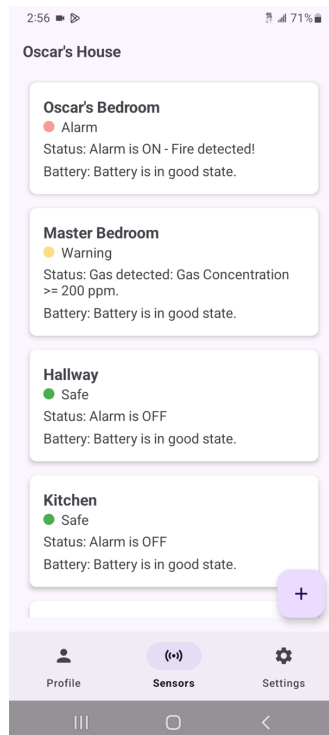
Sensor Name

WiFi Network

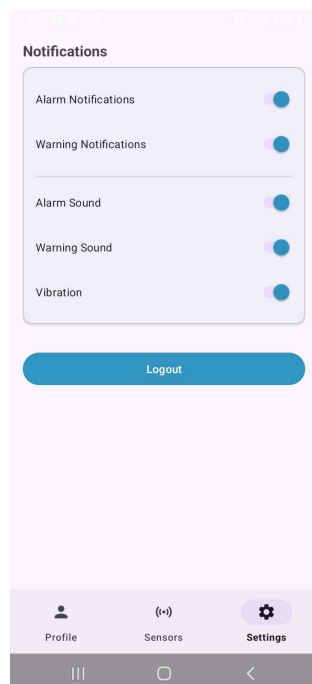
Network Password

Cancel Connect

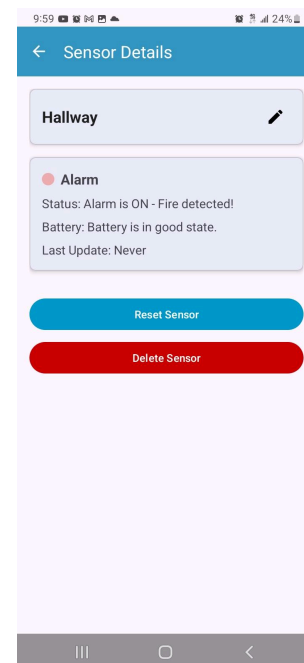
Add New Device



Realtime device monitoring



App configurations



Two-way MCU Communication: Reset/Delete

FSR VALIDATION

Test Name	Successful Criteria	Description	Status	Responsible Engineers
Sensor Functionality	Sensors detect smoke/flame at expected ranges and output a signal.	Ionization and photoelectric sensors turn on; detects smoke on contact; output signals. IR sensor turns on; detects flame at 10ft away; outputs signal. UV sensor turns on; detects flame at 10ft away; outputs signal.	Tested	Martin Ford
Speaker Functionality	Speaker emits sound from signal at >65 dB.		Tested	Martin Ford
Operating Time	System operates and receives full power continuously for 1 hour on each power source.	System operates continuously on hardwired power for 1 hour. System operates continuously on battery power for 1 hour.	Tested	Martin Ford
Network	MCU is able to aggregate data from sensors to router.		Tested	Oscar Hernandez
Dimensions	Mass: System (including casing) does not exceed 0.7lbs. Volume Envelope: System (including casing) does not exceed 6" diameter and 3" in height. Mounting: System casing successfully inserts and holds in ceiling smoke alarm slot.		Tested	Martin Ford
Voltage Levels	Peripherals: Ionization sensor receives 5V; audio speaker receives 4V; photoelectric, IR, and UV sensors each receive 3.3V. Microcontroller receives 3.3V.		Tested	Martin Ford

Thermal Resistance	System operates successfully in environment within temperatures ranging from 40F to 120F		Tested	Martin Ford
Trouble Signal	MCU sends trouble signal to app for malfunctions.		Tested	Oscar Hernandez
User Profile Creation	User is able to create an account and be able to sign in.		Tested	Alfonso Wong
Push Notifications	Be able to push notifications to users and send text to central dispatcher.		Tested	Alfonso Wong
Design UI	Have a working UI and test with emulator.		Tested	Alfonso Wong
Integrate Cloud Service	Mobile app has ability to pull data from cloud service and display it.		Tested	Alfonso Wong
App compatible on Android devices	App can be used on most Android devices		Tested	Alfonso Wong
MCU Peripherals Interface	MCU reads signals from all sensors, and activates/deactivates alarm.		Tested	Martin Ford & Oscar Hernandez
Device and Application Communication	A new device can be added to user profile in application.		Tested	Alfonso Wong & Oscar Hernandez
Real Time Communication	MCU is feeding constant information to database and it is being displayed in real time in mobile application.		Tested	Alfonso Wong & Oscar Hernandez

Trigger Warning from MC	MCU must send a trigger alert of ongoing event and application must send notification to user and dispatch		Tested	Alfonso Wong & Oscar Hernandez
Early Fire Detection	The system reduces the propagation of a fire by minimizing false alarms and notifying the user or fire department as soon as a fire is detected.	System detects smoke/flame with 95% accuracy. Detected alarm activates buzzer.	Tested	All